

Oral L-citrulline supplementation improves erection hardness in men with mild erectile dysfunction.

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OBJECTIVES: To test the efficacy and safety of oral L-citrulline supplementation in improving erection hardness in patients with mild erectile dysfunction (ED). L-arginine supplementation improves nitric oxide-mediated vasodilation and endothelial function; however, oral administration has been hampered by extensive presystemic metabolism. In contrast, L-citrulline escapes presystemic metabolism and is converted to L-arginine, thus setting the rationale for oral L-citrulline supplementation as a donor for the L-arginine/nitric oxide pathway of penile erection.

METHODS: In the present single-blind study, men with mild ED (erection hardness score of 3) received a placebo for 1 month and L-citrulline, 1.5 g/d, for another month. The erection hardness score, number of intercourses per month, treatment satisfaction, and adverse events were recorded.

RESULTS: A total of 24 patients, mean age 56.5 ± 9.8 years, were entered and concluded the study without adverse events. The improvement in the erection hardness score from 3 (mild ED) to 4 (normal erectile function) occurred in 2 (8.3%) of the 24 men when taking placebo and 12 (50%) of the 24 men when taking L-citrulline ($P < .01$). The mean number of intercourses per month increased from 1.37 ± 0.93 at baseline to 1.53 ± 1.00 at the end of the placebo phase ($P = .57$) and 2.3 ± 1.37 at the end of the treatment phase ($P < .01$). All patients reporting an erection hardness score improvement from 3 to 4 reported being very satisfied. **CONCLUSIONS:** Although less effective than phosphodiesterase type-5 enzyme inhibitors, at least in the short term, L-citrulline supplementation has been proved to be safe and psychologically well accepted by patients. Its role as an alternative treatment for mild to moderate ED, particularly in patients with a psychologically fear of phosphodiesterase type-5 enzyme inhibitors, deserves further research. Copyright © 2011 Elsevier Inc. All rights reserved.